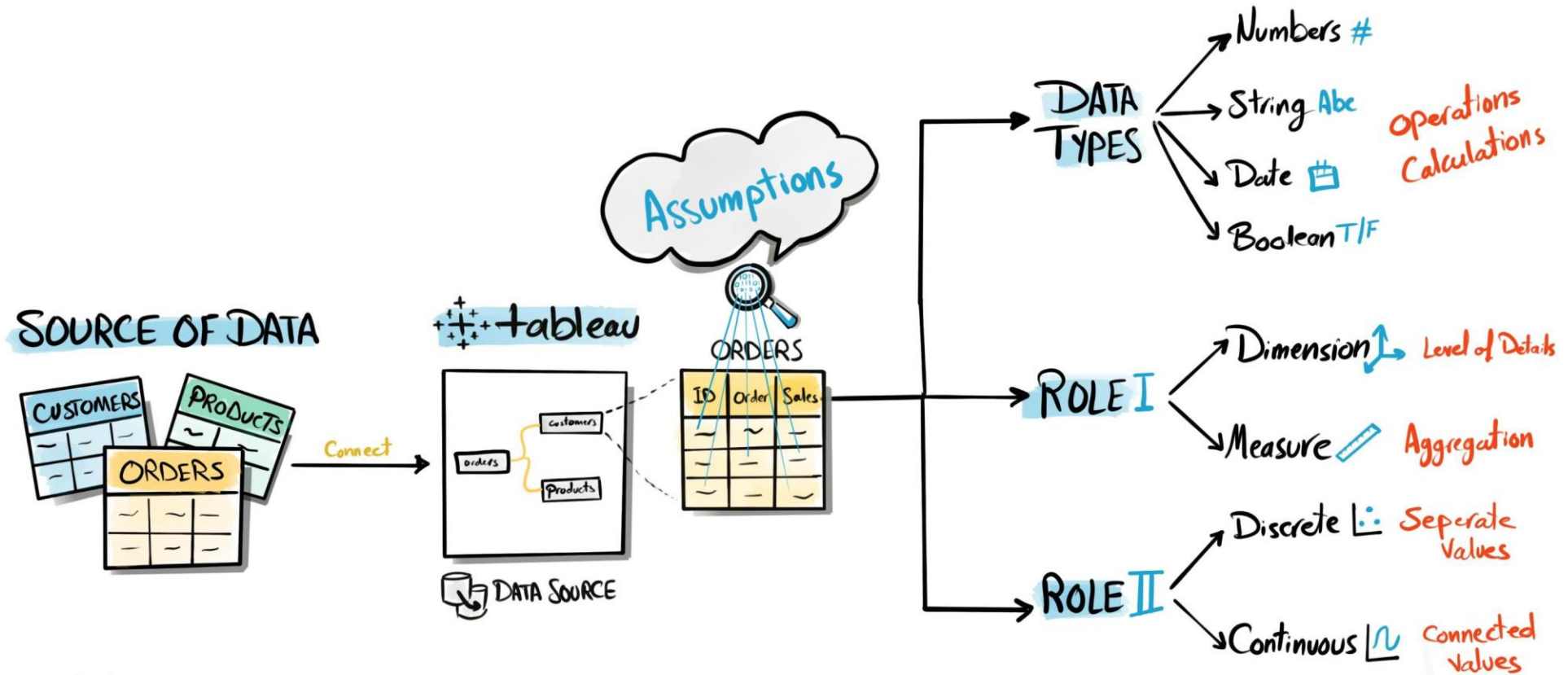
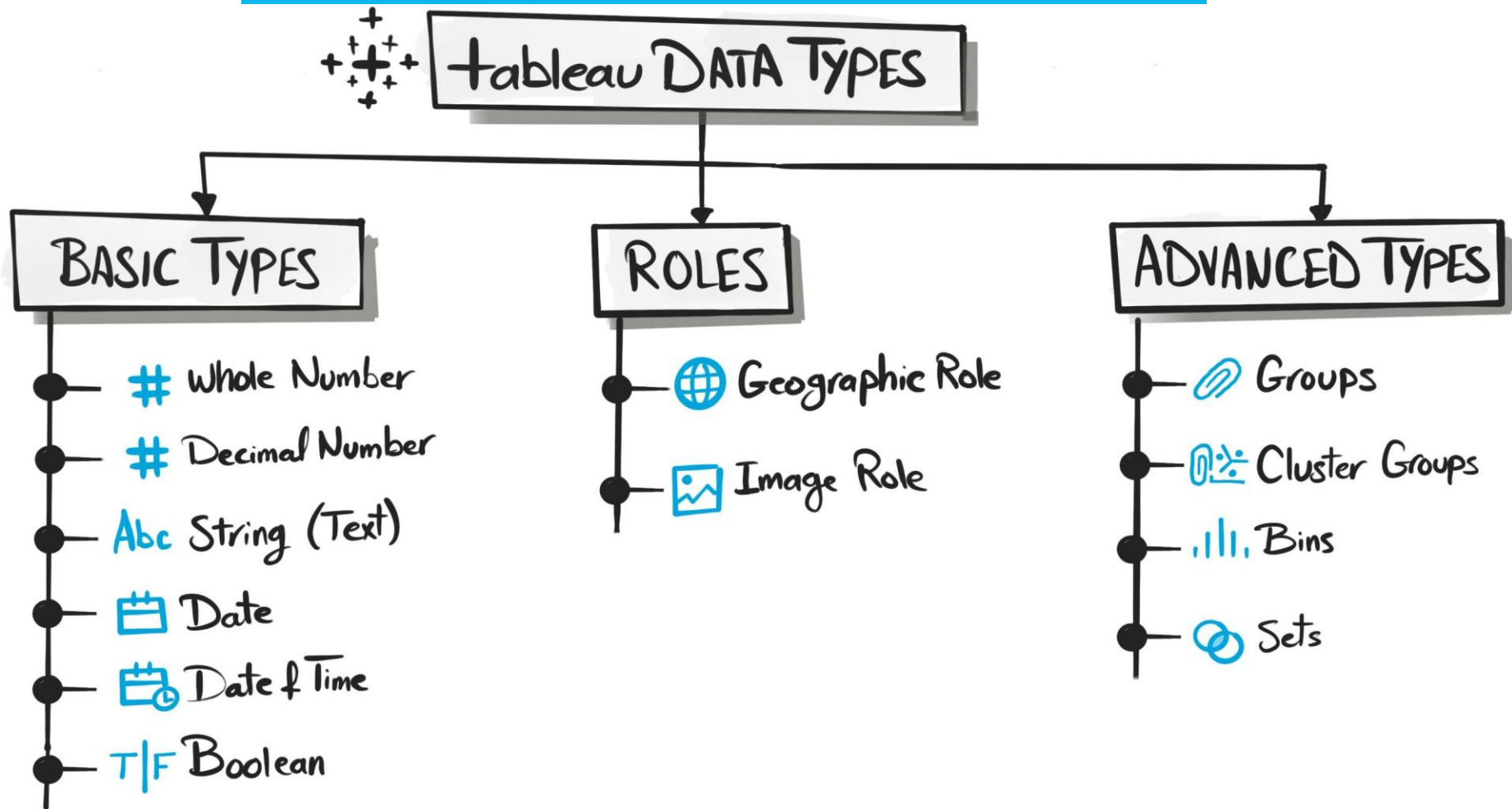


# Tableau Metadata



# Tableau Data Types

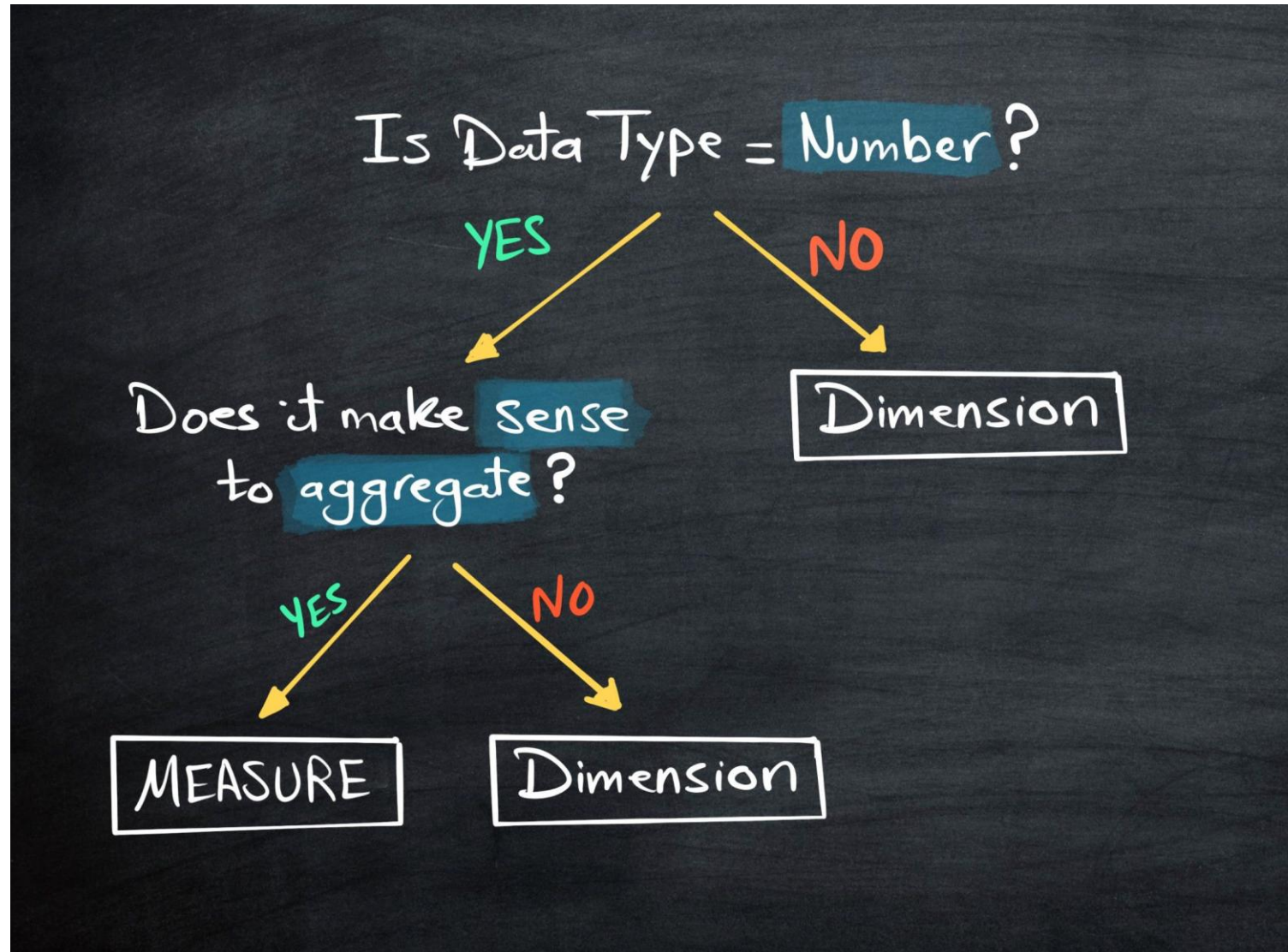


# Basic Data Types

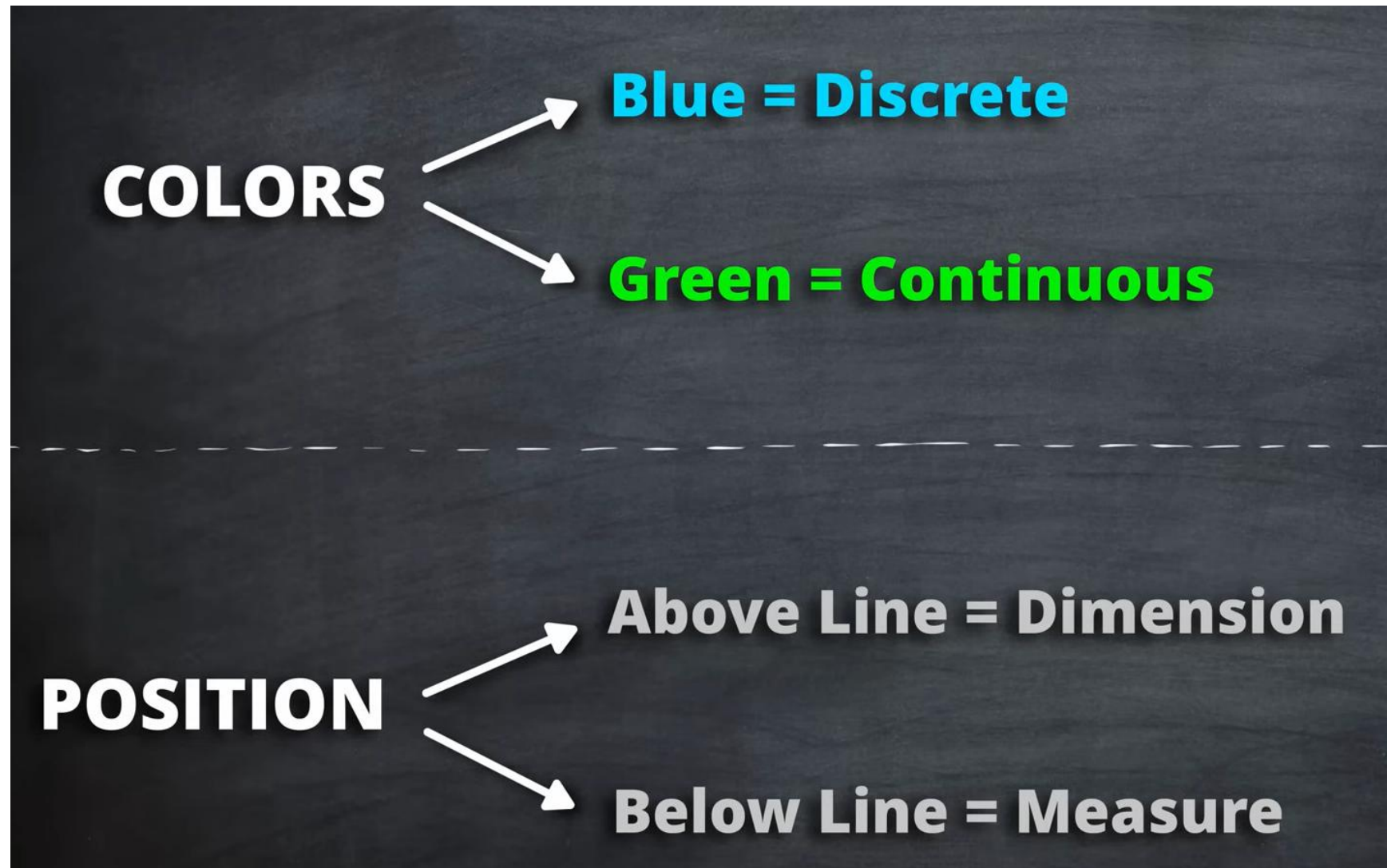
## BASIC TYPES

|   |     |                  |                     |
|---|-----|------------------|---------------------|
| ● | #   | Number (Whole)   | 1234                |
| ● | #   | Number (Decimal) | 35.98               |
| ● | Abe | String (Text)    | Hello World!        |
| ● | 📅   | Date             | 20/08/2025          |
| ● | 📅⌚  | Date & Time      | 20/08/2025 18:20:58 |
| ● | T/F | Boolean          | TRUE, FALSE         |

## Dimensions vs Measures







# Dimensions vs Measures

## SECRET FORMULA

**FORMULA:** Measure By Dimension

**EXAMPLES:** Sales By Product

Profit By Category

Quantity By Country



## DIMENSION

|                |
|----------------|
| Customer_ID    |
| First_Name     |
| Last_Name      |
| Postal_Code    |
| City           |
| Country        |
| Customer_Email |
| Age            |
| Order_ID       |
| Customer_ID    |
| Product_ID     |
| Order_Date     |
| Shipping_Date  |
| Product_ID     |
| Product_Name   |
| Product_Image  |
| Category       |



## MEASURE

|                   |
|-------------------|
| Sales             |
| Quantity          |
| Discount          |
| Profit            |
| Unit_Price        |
| Score             |
| Count (Customers) |
| Count (Orders)    |
| Count (Products)  |

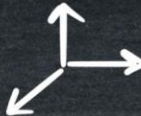


## VIZ

|                           |
|---------------------------|
| Sales by Order Date       |
| Sales by Shipping Date    |
| Sales by Country          |
| Sales by City             |
| Sales by Age              |
| Sales by Category         |
| Sales by Product Name     |
| Profit by Order Date      |
| Profit by Shipping Date   |
| Profit by Country         |
| Profit by City            |
| Profit by Age             |
| Profit by Category        |
| Profit by Product Name    |
| Quantity by Order Date    |
| Quantity by Shipping Date |
| Quantity by Country       |
| Quantity by City          |
| ...                       |



## Dimensions vs Measures

| <br>tableau | Dimensions  | Measures  |
|----------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------|
| Definition                                                                                   | Descriptive Values                                                                             | Quantitative Values                                                                          |
| Example                                                                                      | Category, Country, ID                                                                          | Sales, Profit, Quantity                                                                      |
| Aggregation                                                                                  | Can <b>Not</b> be aggregated                                                                   | Aggregable [SUM, AVG, MIN, MAX]                                                              |
| Data Types                                                                                   | [String, Date, Boolean, Number]<br><small>Abe      ☰      TIF      #</small>                   | [Number]<br><small>#</small>                                                                 |
| Role of Analysis                                                                             | Filtering, Grouping, Organising Data                                                           | Calculations & Numerical Analysis                                                            |
| Granularity                                                                                  | LOD: Level of Details                                                                          | Determine Quantity being Measured                                                            |

## Discrete vs Continuous

**Discrete**

11 Values

0  
1  
2  
3  
4  
5  
6  
7  
8  
9  
10

**Continuous**

$\infty$  Values

1  
1.11  
1.12  
1.13  
1.14  
1.15  
:  
:  
:  
2



# Discrete vs Continuous

## Discrete

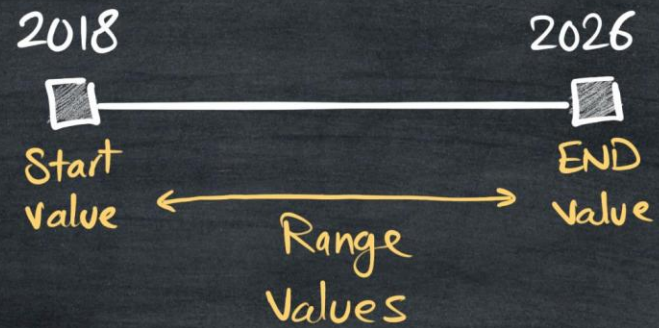
Filter-Category

- ☐ All
- ☐ Art
- ☐ Copiers
- ☐ Storage
- ☐ Tables
- ☐ ~~~

Distinct  
Values

## Continuous

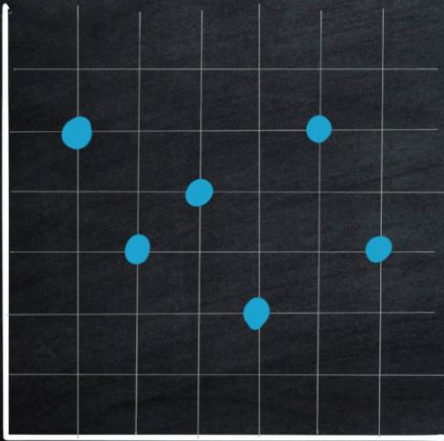
Filter-Date



# Discrete vs Continuous

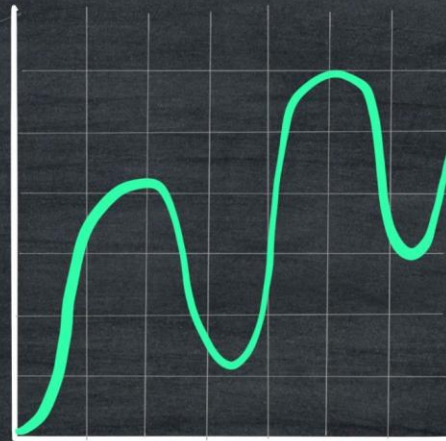
## Discrete

Disconnected & Separated  
Values



## Continuous

Connected & unbroken chain  
of Values





### Discrete

#### Many Sorting options

- Ascending
- Descending
- Data Source order
- Alphabetic
- Field
- Manual
- Nested

### Continuous

#### Limited Sorting Options

- Ascending
- Descending



## Discrete vs Continuous

### Discrete

Purpose:

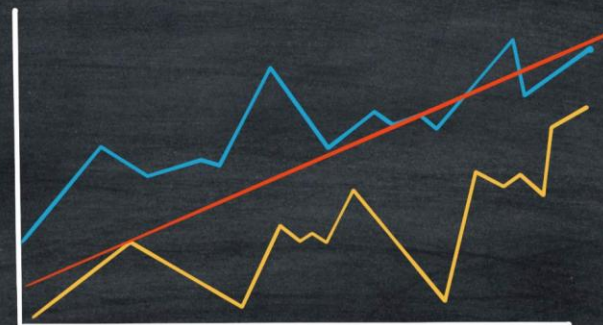
Deep Dive Analysis



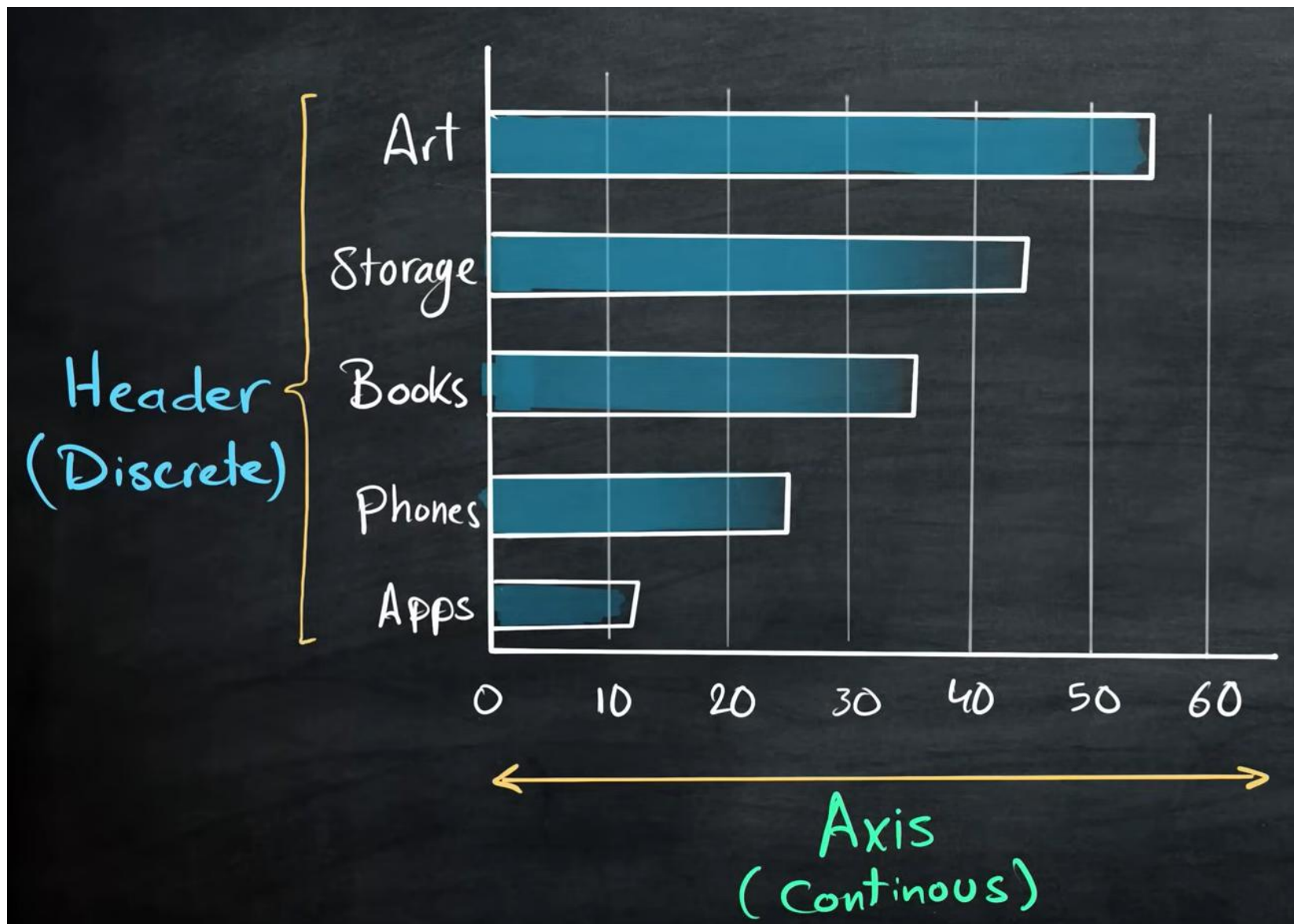
### Continuous

Purpose:

Big Picture Analysis

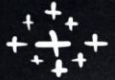


## Discrete vs Continuous



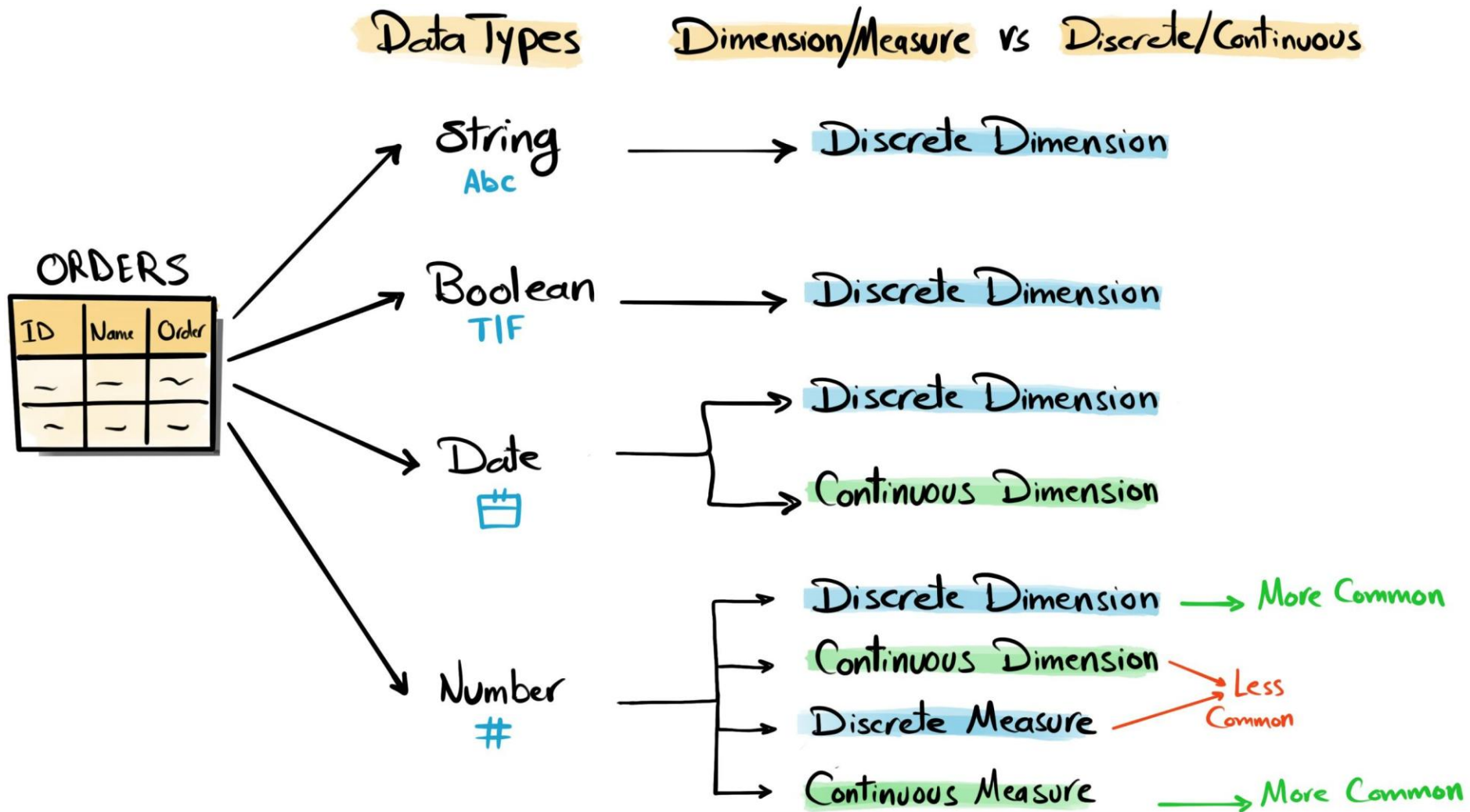


# Discrete vs Continuous

| <br>tableau | Discrete                                                                                                | Continuous                                                                                           |
|----------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------|
| Definition                                                                                   | Disconnected Values  | Connected values  |
| Example                                                                                      | [0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10]                                                                      | [1, 1.1, 1.12, 1.13, 1.14, ..., 2] $\infty$                                                          |
| Colors                                                                                       | Blue Pill            | Green Pill        |
| Filters                                                                                      | Distinct values      | Range value       |
| Views                                                                                        | Header                                                                                                  | Axis                                                                                                 |
| Sorting                                                                                      | Many Sorting options                                                                                    | Limited Sorting Options                                                                              |
| Purpose                                                                                      | Specific Scenario   Deep Dive                                                                           | Big Picture   Trends Analysis                                                                        |

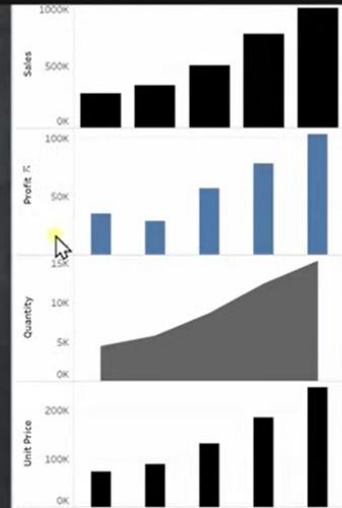


# Tableau Metadata

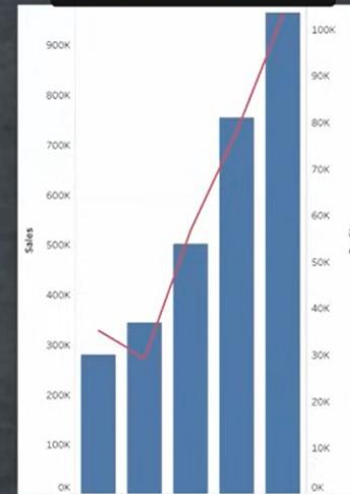


# Tableau Charts

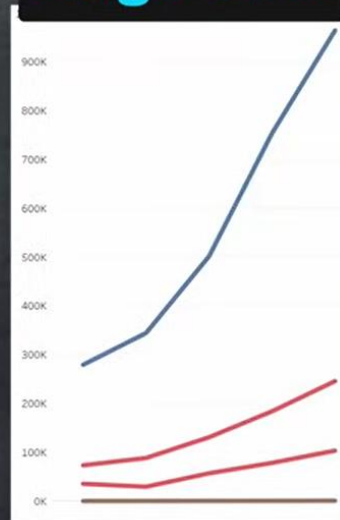
## Individual Axes



## Dual Axes



## Single Axes



Add multiple measures

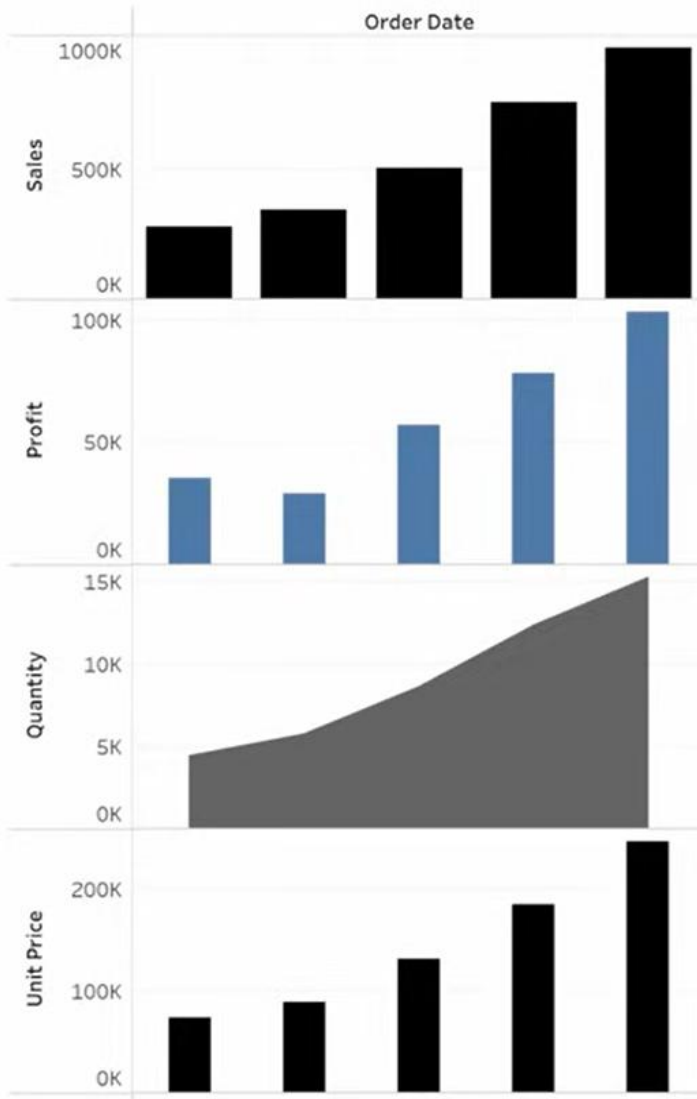


- + Customize Charts Individually
- + Unlimmited Nr. of Measures
- Hard to **Compare** Measures

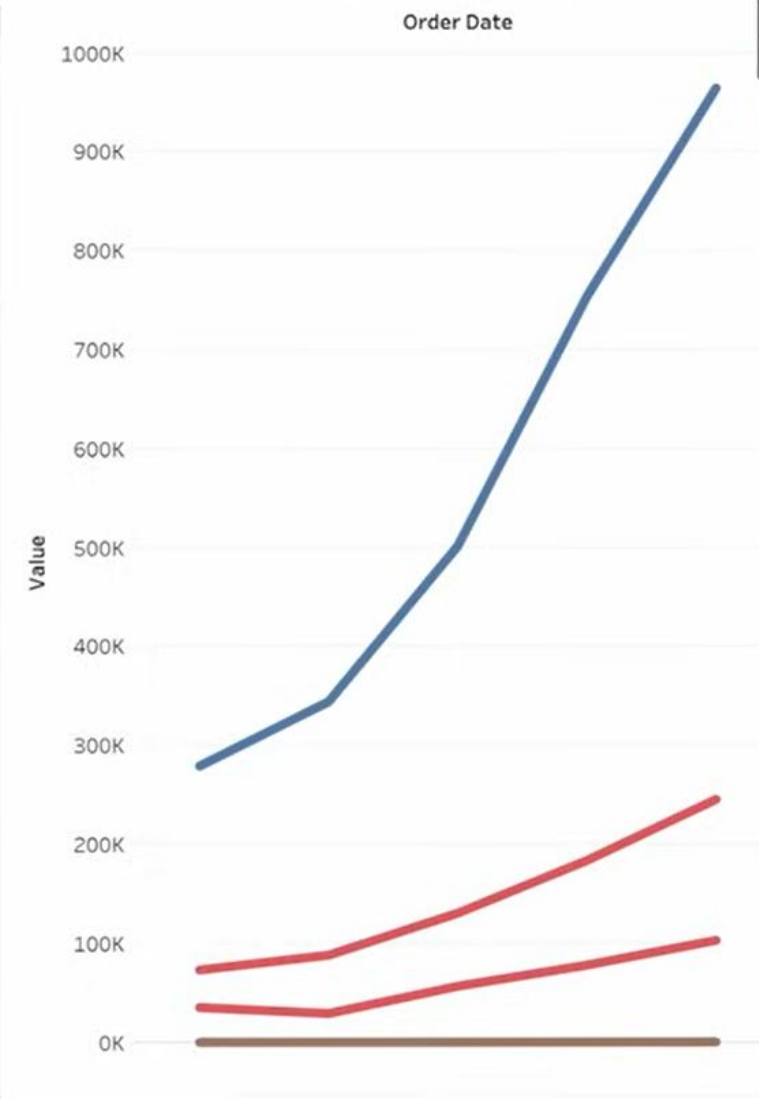
- + Easy to Compare Measures
- + Unlimmited Nr. of Measures
- Sharing **Same** Visual Configuration

- + Easy to Compare Measures
- + Customize Charts Individually
- + Option of Sync/Async Axes
- Only **Two** Measures

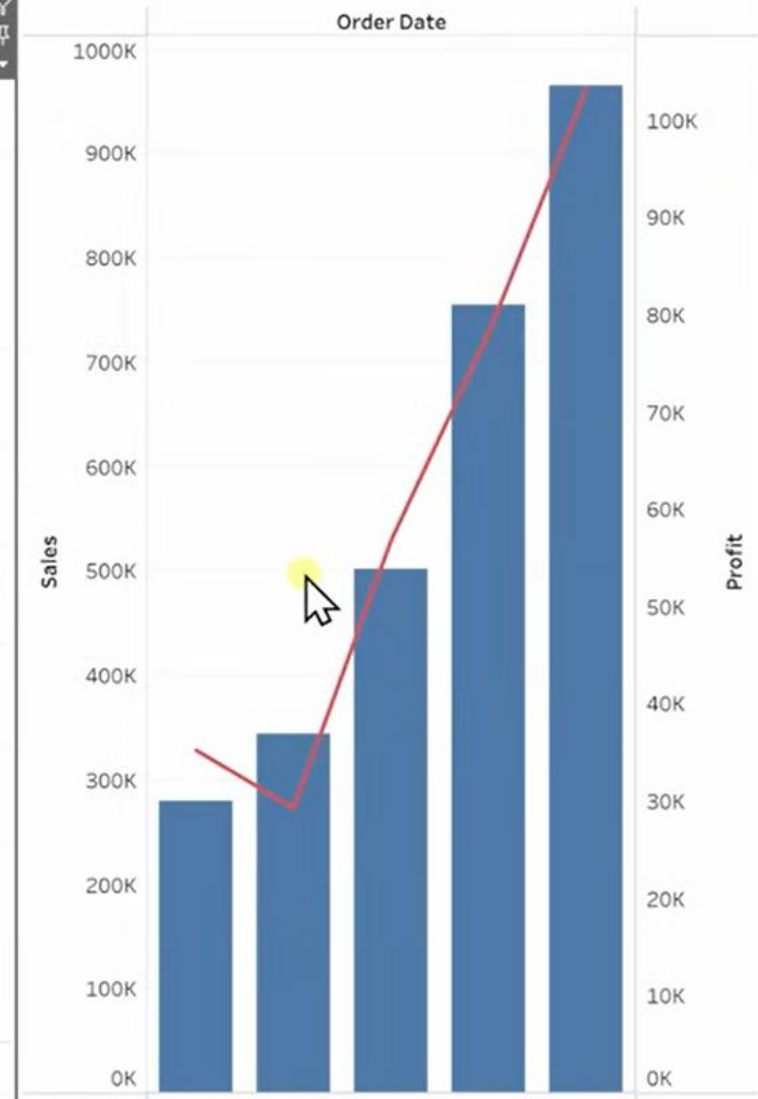
Individual Axis



Single Axis (Measure Names & Values)



Dual Axis



# Tableau Charts

Let us create  
28 charts

# Tableau Charts

1. Bar chart (row)  
**Use case:** ranking. Use it with HIGH cardinality dimensions
2. Bar chart (column)  
**Use case:** ranking. Use it with LOW cardinality dimensions
3. Bar chart (side-by-side)  
**Use case:** compare multiple values across different categories
4. Bar chart (over time)  
**Use case:** analyze changes over time to compare data across different periods, analyze trends
5. Stacked bar chart  
**Use case:** 1) analyze change over time. 2) part-to-whole relationships



# Tableau Charts

6. Full (100%) stacked bar chart

**Use case:** part-to-whole relationships

7. Multiple small bar charts

**Use case:** part-to-whole relationships

8. Bar-in-bar chart

**Use case:** correlation. Compare two measures using dual axis

9. Bar code charts

**Use case:** compare the distribution of the values of a high cardinality dimension within another dimension

10. Line chart

**Use case:** analyze change over time

# Tableau Charts

11. Customized line chart

12. Multiple line charts

**Use case:** compare the values of one dimension over time

13. Dual-line chart

**Use case:** compare measure values over time in one view

14. Cumulative line chart

**Use case:** show the development and change in growth over time

15. Multiple small line charts

**Use case:** analyze change over time

16. Highlighted line chart (1)

17. Highlighted line chart (1)

# Tableau Charts

18. Scatter plot

**Use case:** correlation. Compare two measures. Find outliers

19. Customized scatter plot

20. Pie chart

**Use case:** part-to-whole relationships

21. Heat map

**Use case:** correlation between two categories

22. Basic map

23. Map with a symbol

24. Histogram (1 measure)

**Use case:** distribution of data

25. Histogram (2 measures)



# Tableau Charts

26. Box plot

**Use case:** understand the data distribution of the dataset

27. KPI chart

**Use case:** analyze the performance of your business

21. BANS

# Tableau Charts

How to choose the  
right chart?

# Tableau Charts

## CHANGE OVER TIME

Show trends of **patterns** over a **continuous period**

### Questions

- How does the data change evolve **over time**?
- Are there any trends or patterns in data **over time**?

### LINE CHART



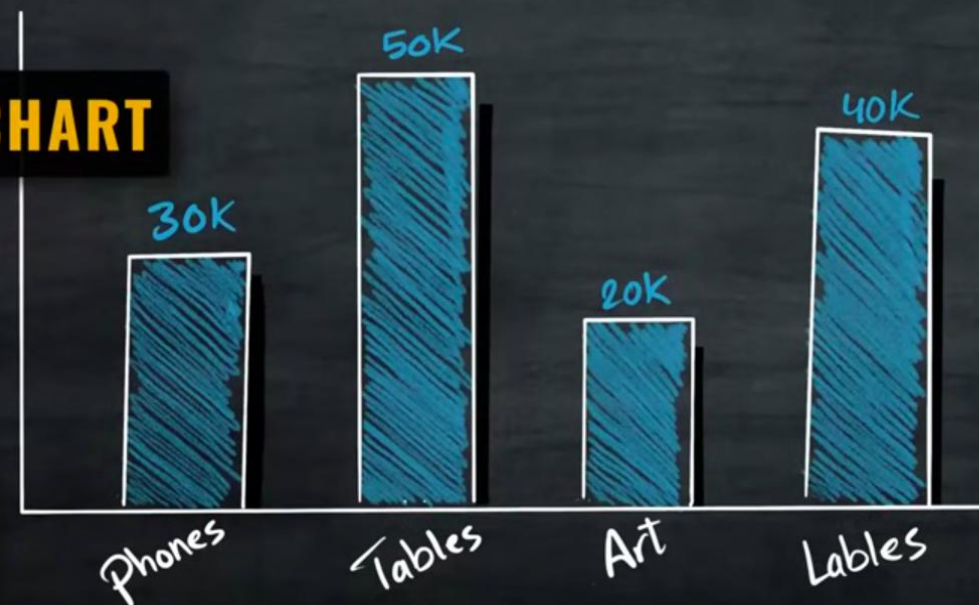
## MAGNITUDE

Show size **comparisons**. These can be relative absolute

### Examples

- Which categories have the highest and lowest sales
- Show **comparison** of sales across different categories

### BAR CHART





# Tableau Charts

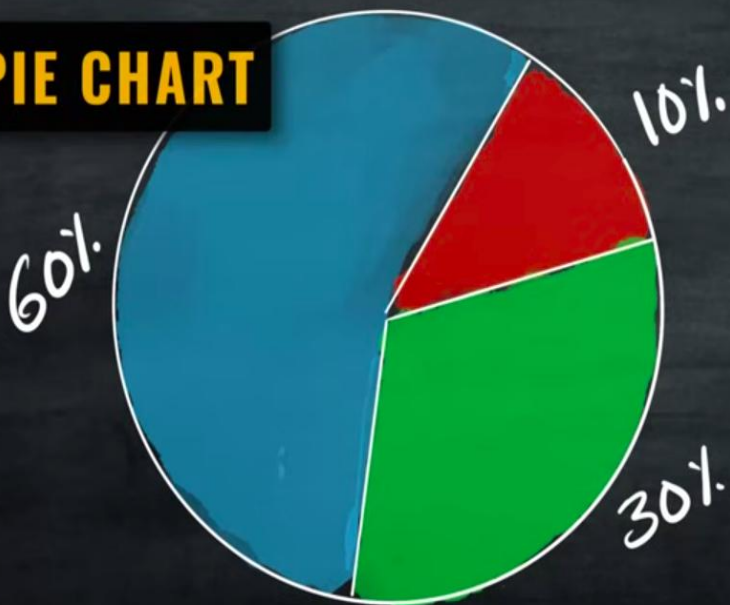
## PART-TO-WHOLE

Show how a **whole** breaks down into its **components**

### Question

- How much each category **contributes** to **overall** sales?

### PIE CHART



## CORRELATION

Show the **relationships** between two or more measures

### Questions

- Is there a **correlation** between two measures?
- How strongly **related** are the two variables?

### SCATTER PLOT



# Tableau Charts

## RANKING

Show an item's **position** in an ordered list (Relative Ranking)

### Examples

- Show **top 10** customers by sales
- Show **lowest 10** products by sales



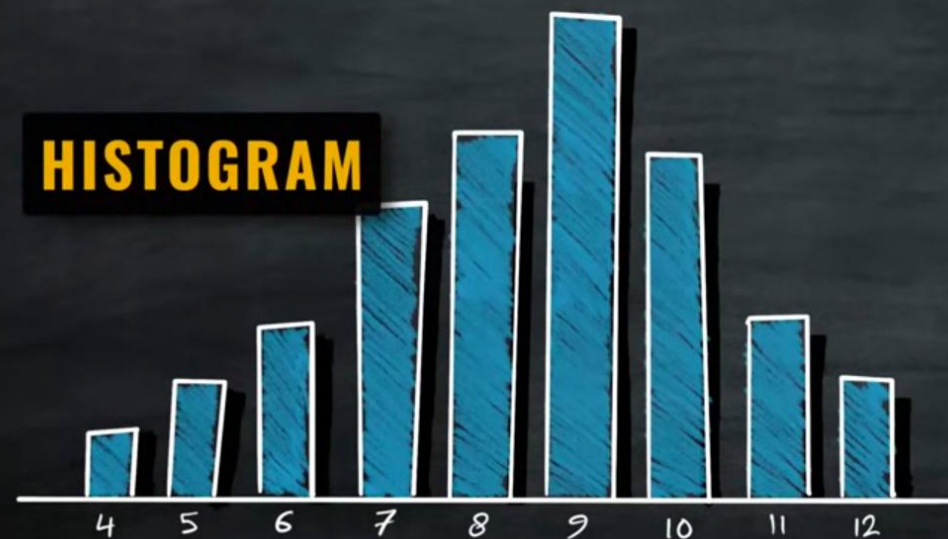
## DISTRIBUTION

Show values in a dataset and how **often they occur**

### Questions

- What is the **distribution** of customers' ages?
- What are the busiest times in our workday?

### HISTOGRAM





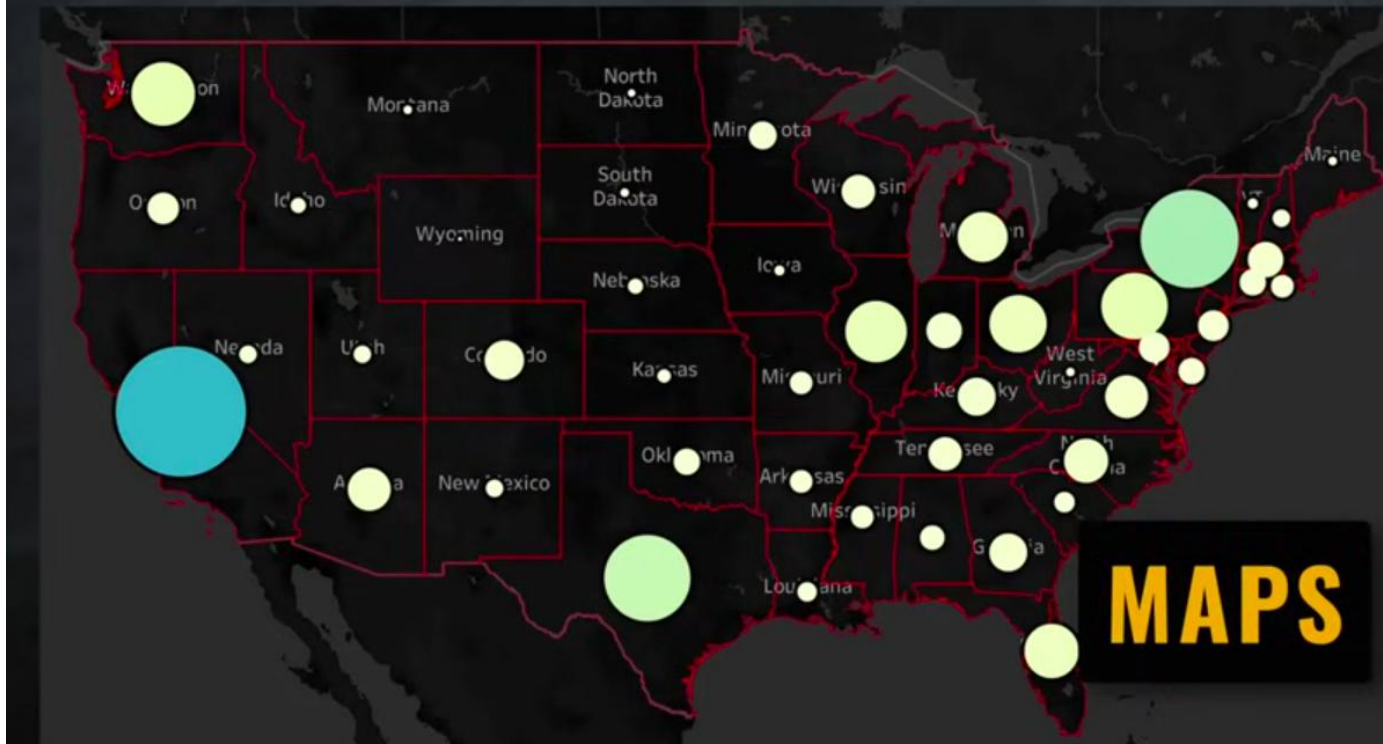
# Tableau Charts

## SPATIAL

Use when **geospatial patterns** in your data are more important than anything else

### Question

- Which **city/state** has the highest sales?



# Containers

## CONTAINERS

---

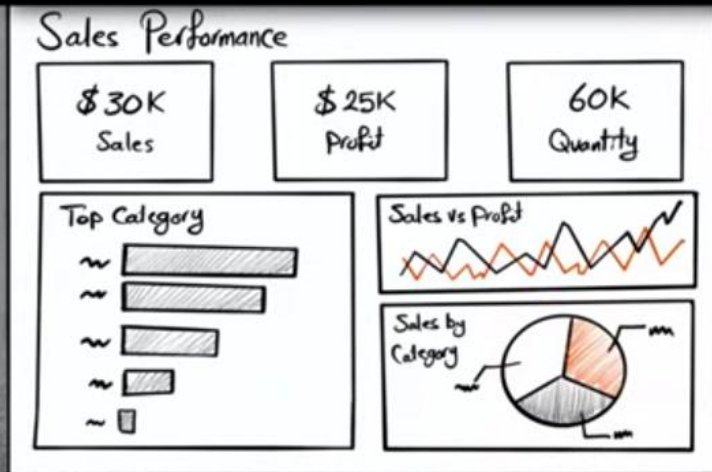
**Group up together** related tableau objects (charts, images, text, ..etc), so you can quickly position them.

Once you change the size, position, or format of the container, **all objects** within the container will adjust **automatically**!

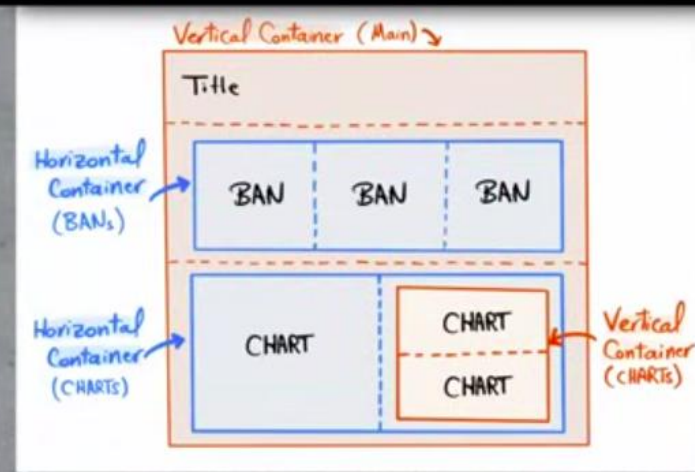


# Dashboard Development Process

## #1 Dashboard Mockup



## #2 Container Mockup



## #3 Build Containers



# Dashboard Mockup

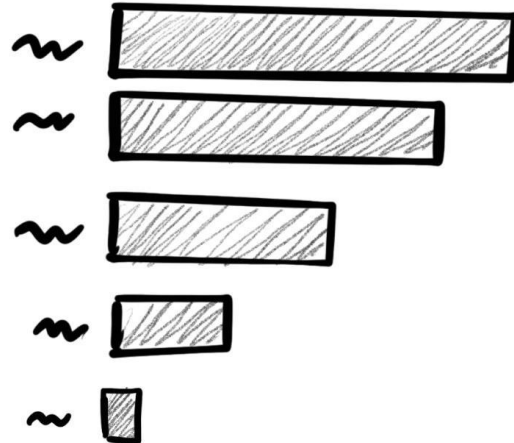
## Sales Performance

\$30K  
Sales

\$25K  
Profit

60K  
Quantity

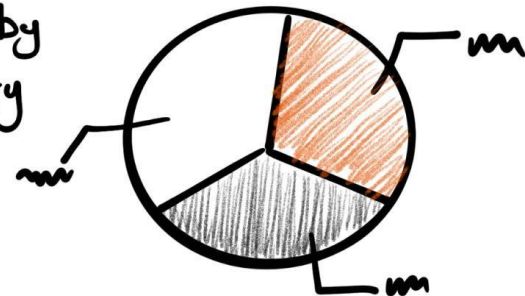
### Top Category



### Sales vs Profit



### Sales by Category



# Container Mockup

